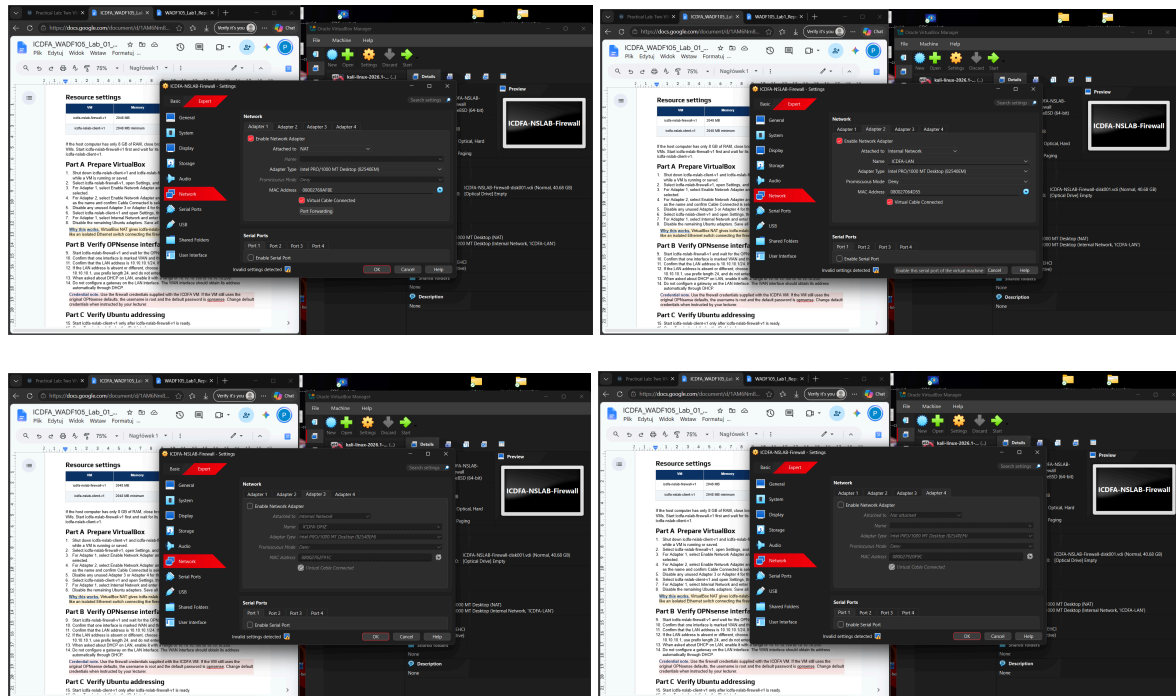
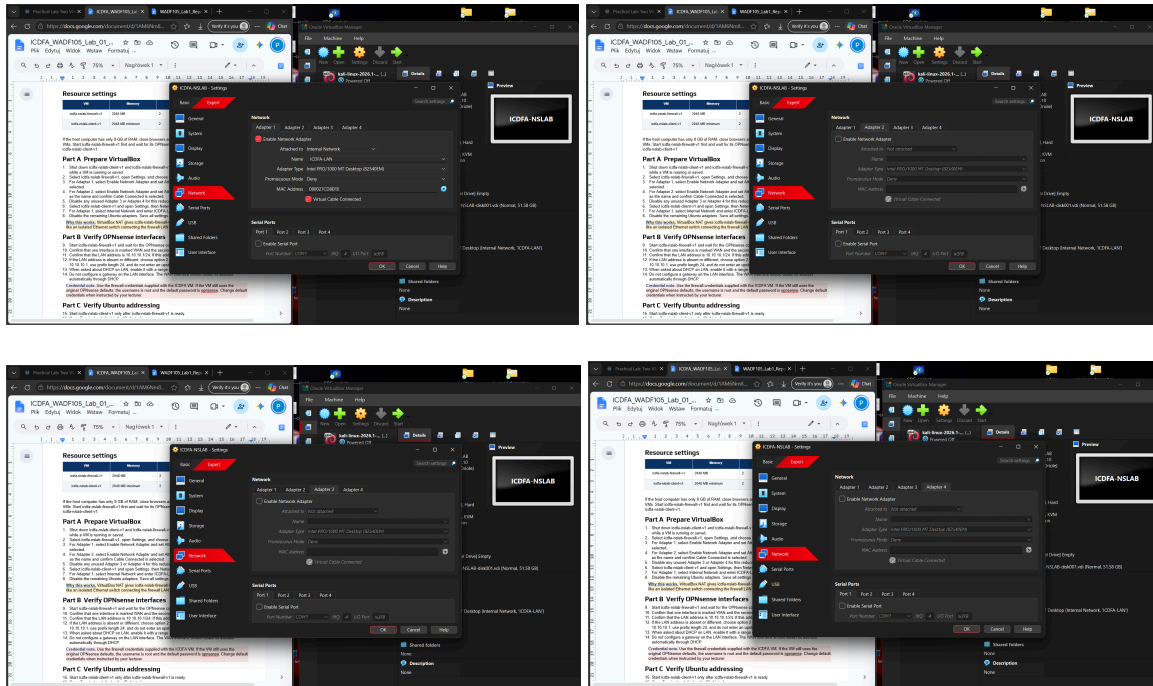
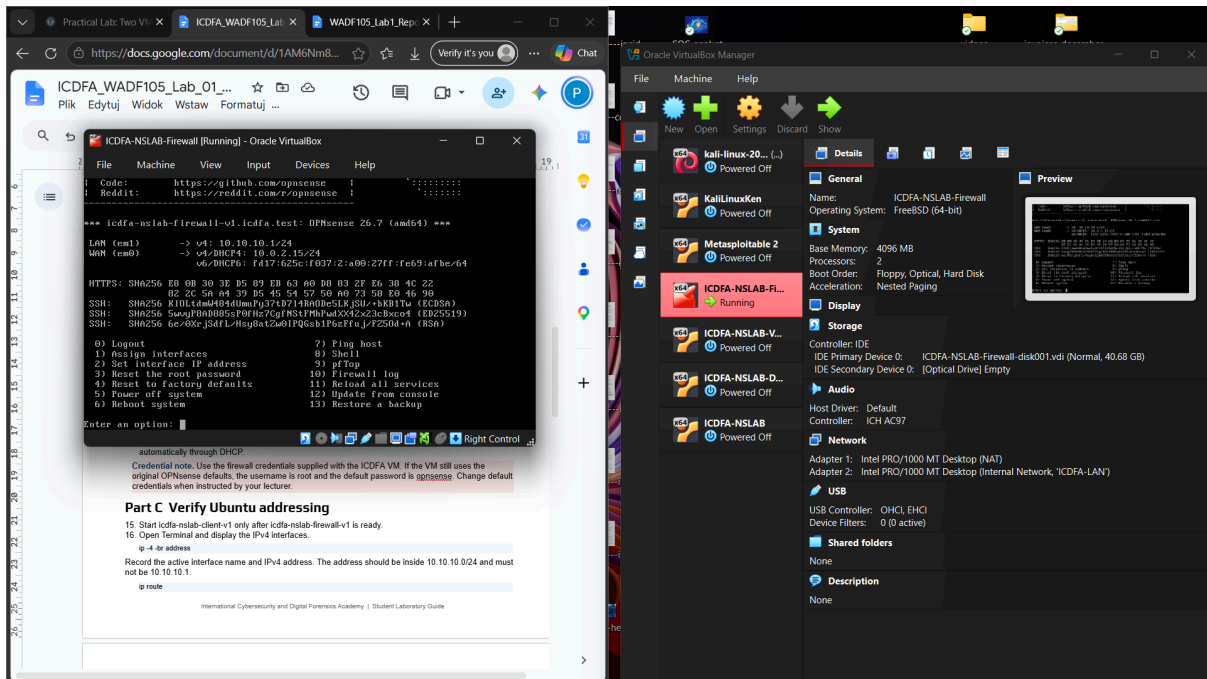




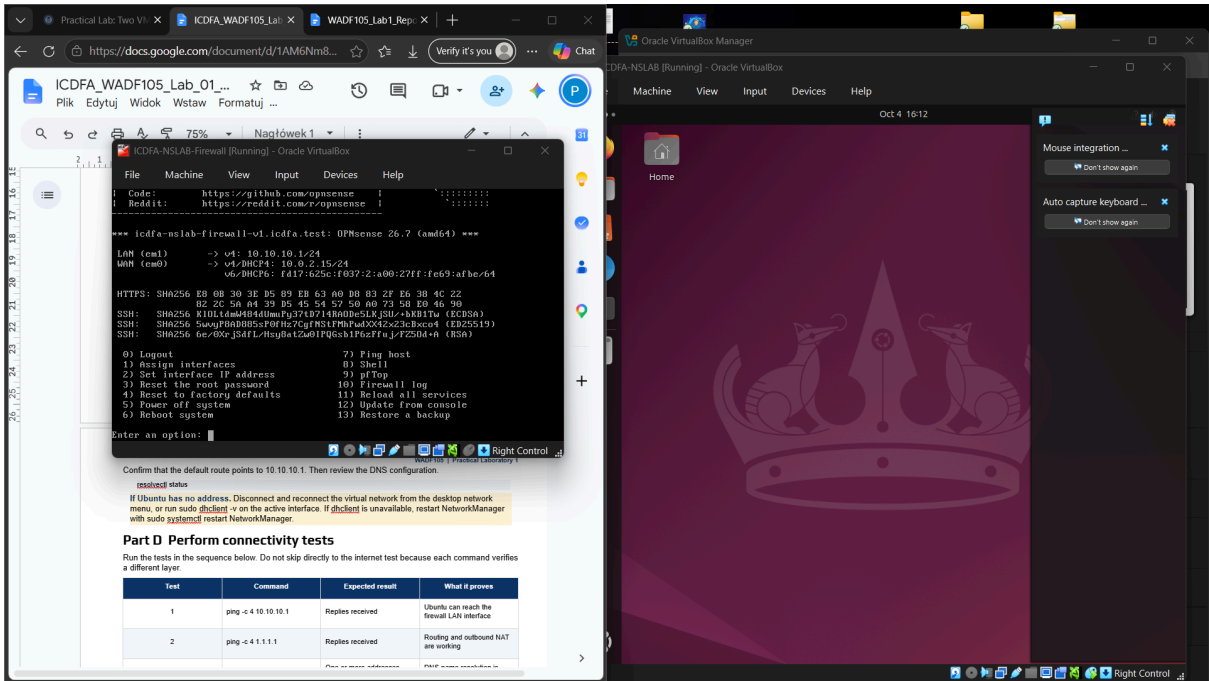
Client adapters settings:



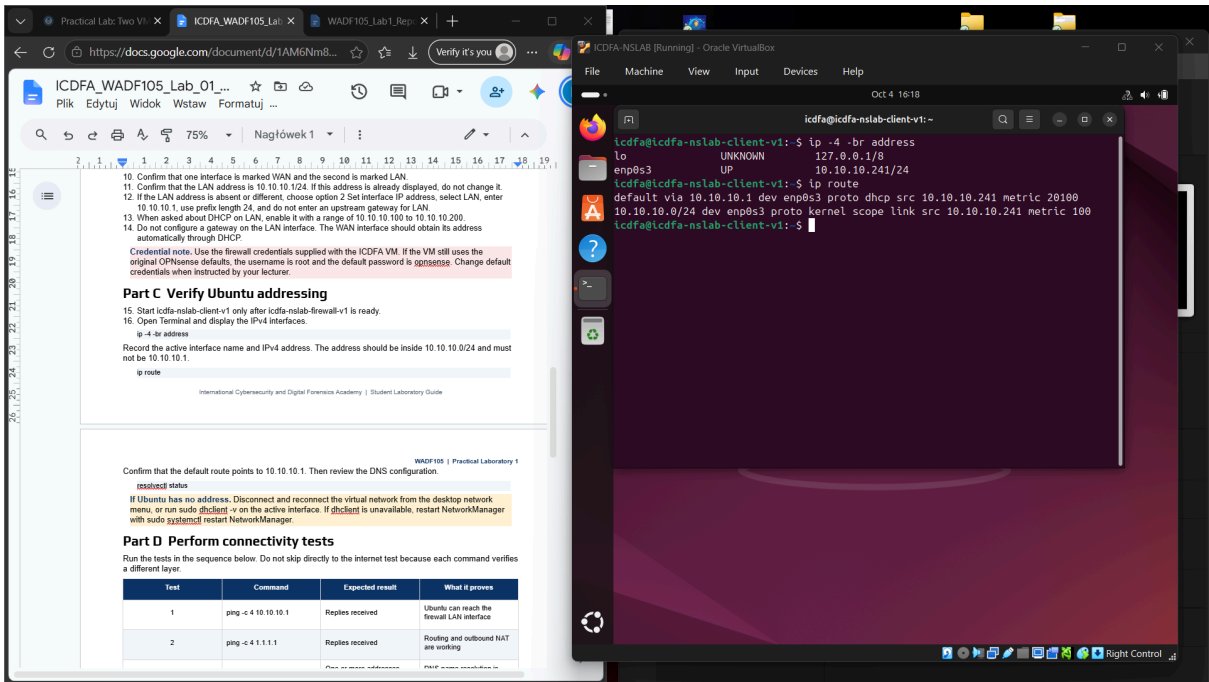
Starting firewall:



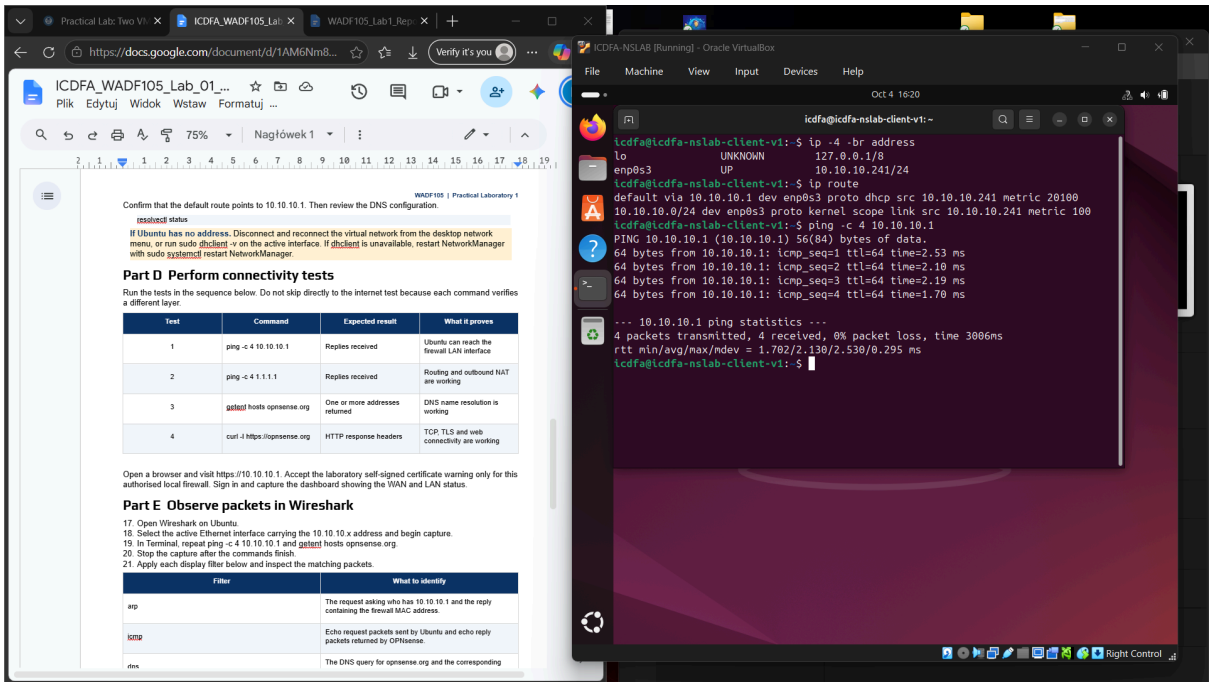
Starting Client/Ubuntu:



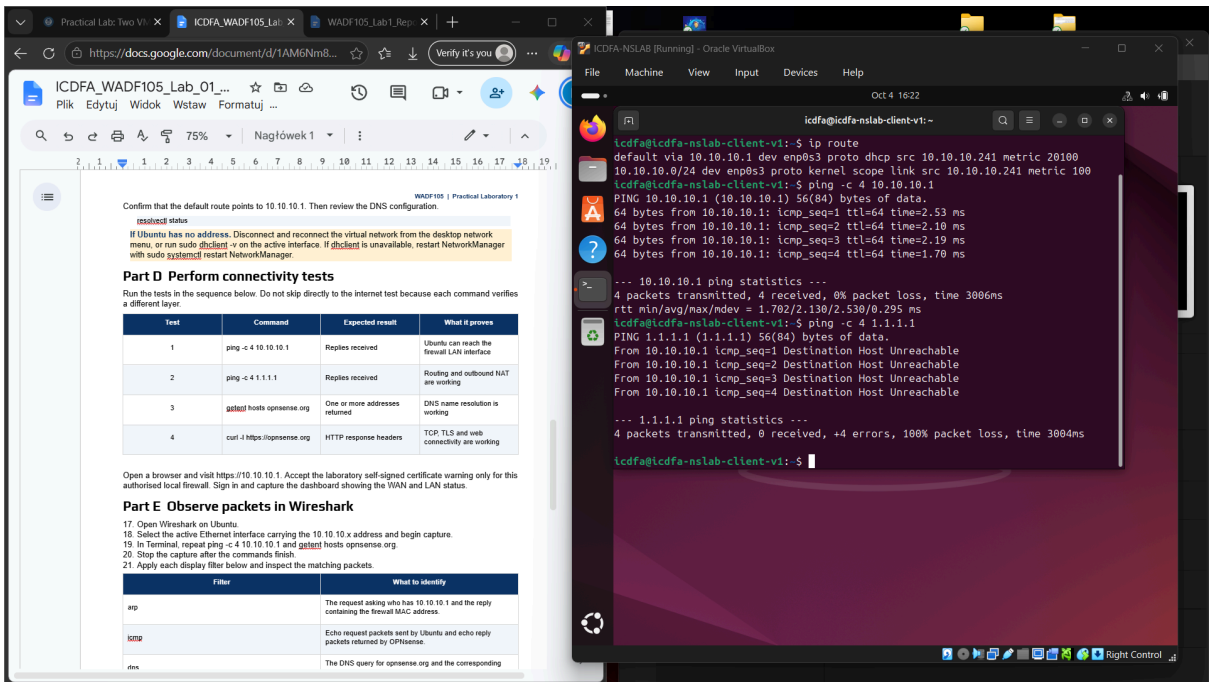
Command “ip -4 -br address”, “ip route” :



Command “ping -c 4 10.10.10.1” :



Command “ping -c 4 1.1.1.1” :



Issue Summary & Resolution

During Lab 1, I encountered a major connectivity issue where every ping command executed from my Ubuntu client machine failed with errors and 100% packet loss when attempting to connect to external IP addresses like 1.1.1.1.

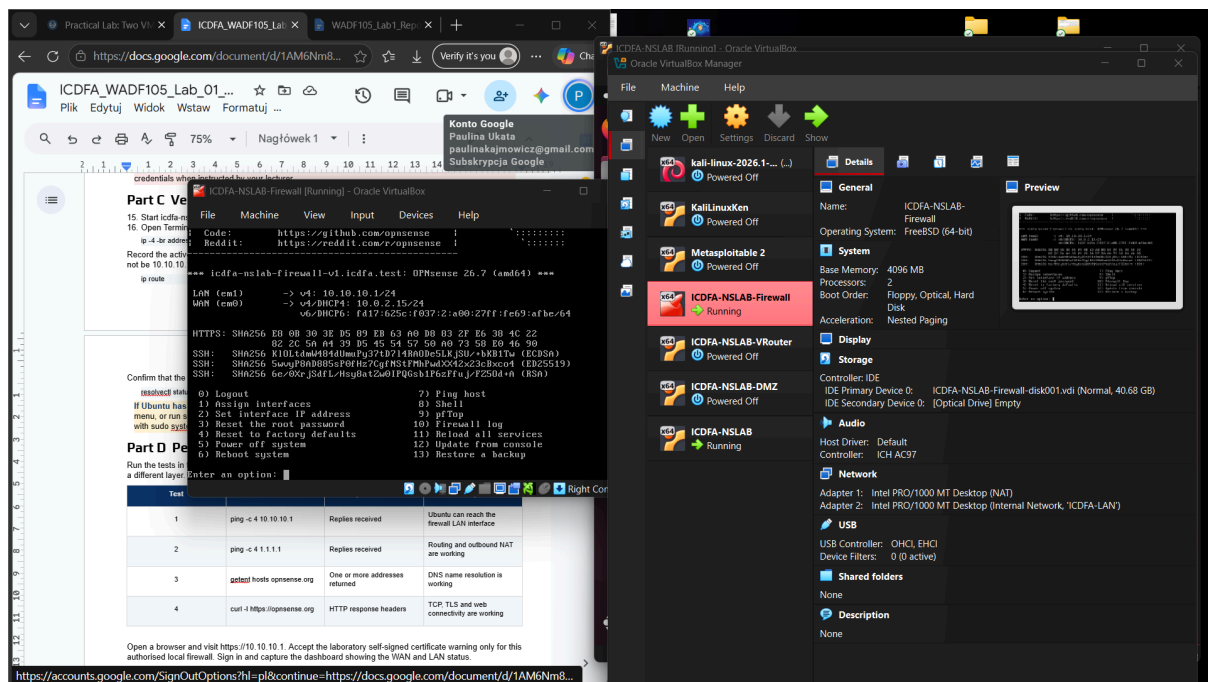
What Went Wrong: The problem occurred in **Part B** because the network interfaces (eth0 / eth1) were completely messed up and misaligned between the LAN and WAN bindings in the firewall script. Because the interface assignments were inverted, the firewall script

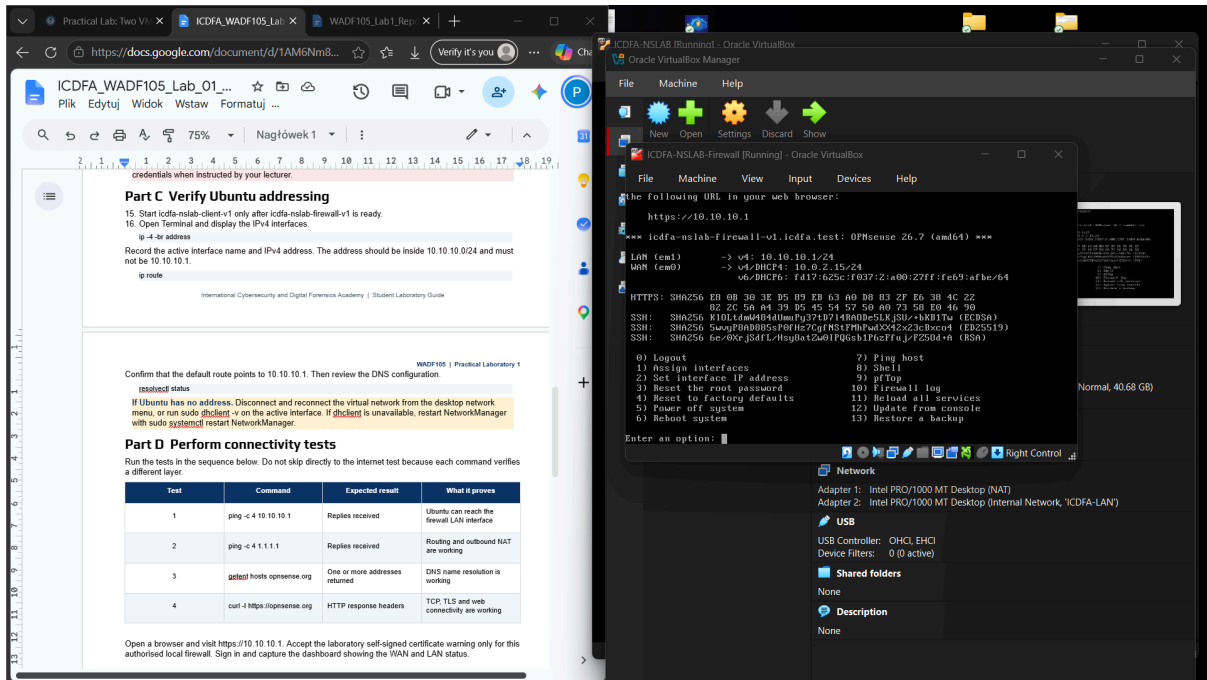
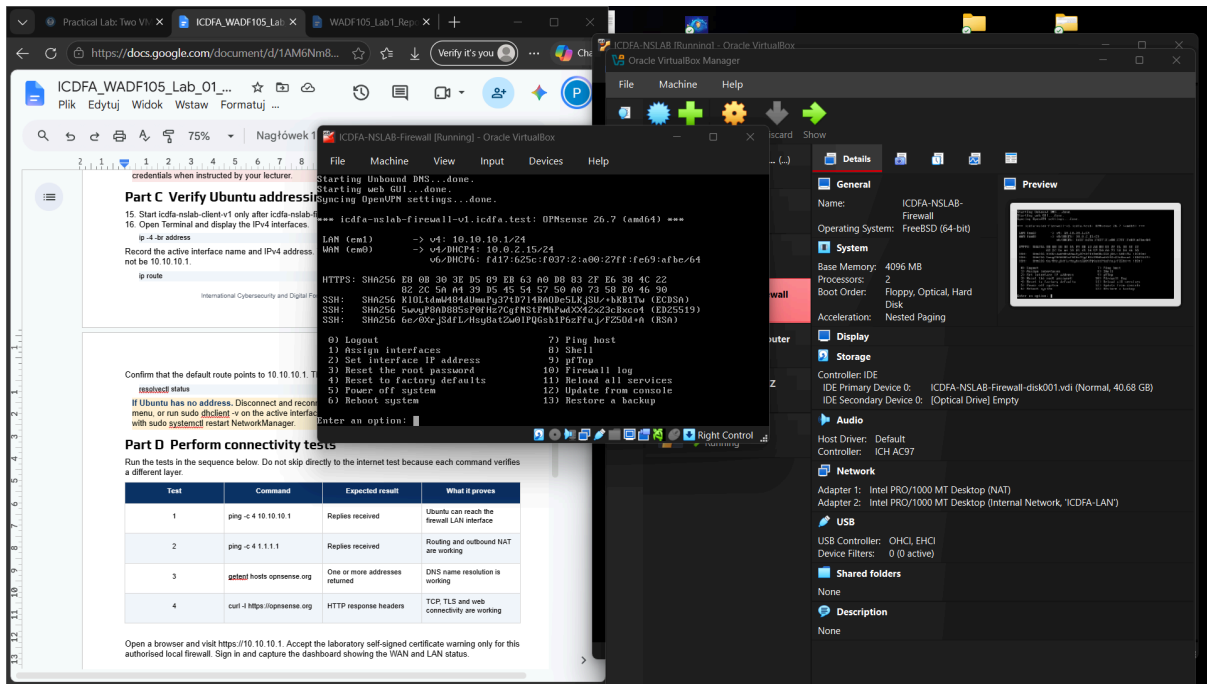
applied rules to the wrong network adapters and blocked all outbound traffic from reaching the internet, causing every ping attempt to fail with errors.

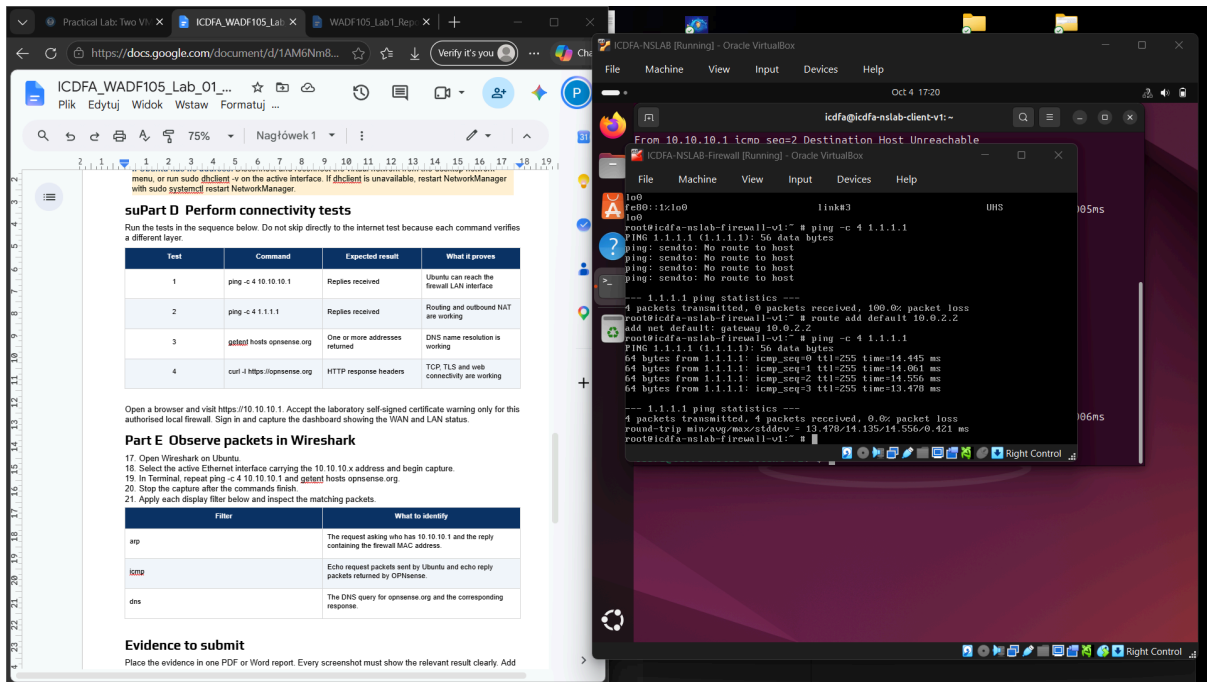
How I Resolved It:

1. First, i tried restarting the network/firewall manager directly from the terminal to see if the service would pick up the correct configuration.
2. When that didn't fix it, i went into the firewall VM settings/console and manually re-assigned the **WAN** and **LAN** interfaces to their correct physical adapters (eth0 / eth1).
3. I then re-ran the firewall script to enforce the corrected interface bindings and clear out the old routing rules.

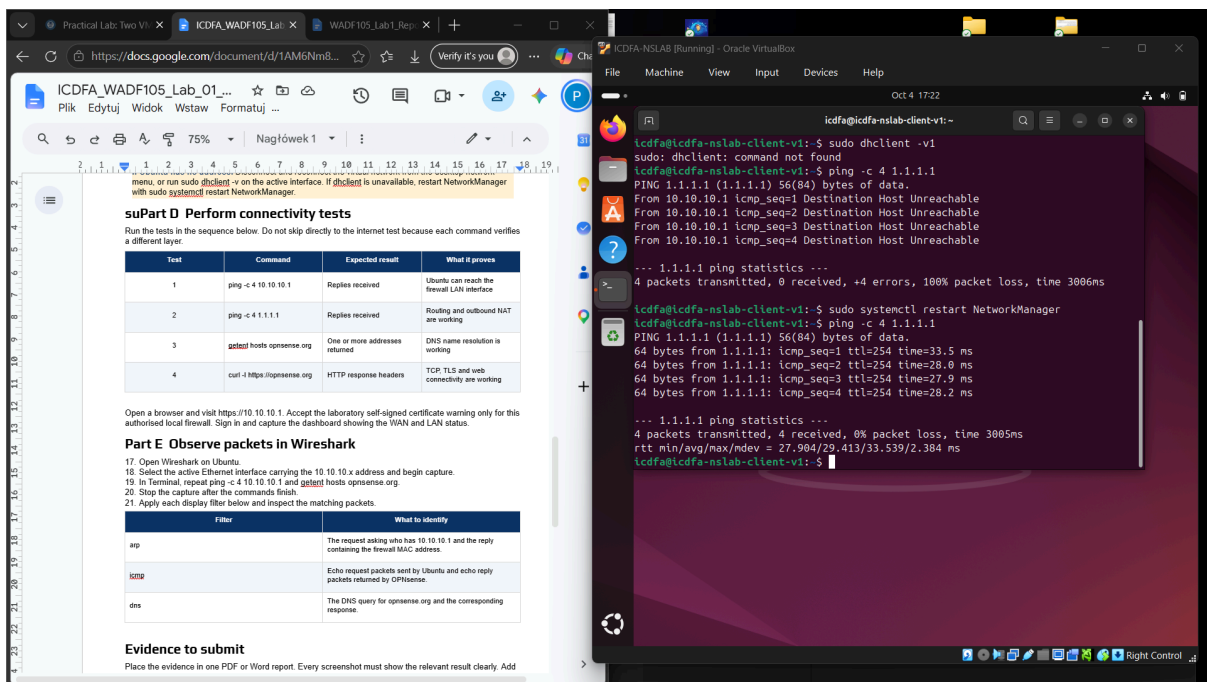
Verification: After re-assigning the interfaces and applying the updated firewall script, I returned to the Ubuntu terminal and tested `ping -c 4 1.1.1.1`. The command succeeded with **0% packet loss**, confirming that traffic was properly routing through the firewall.



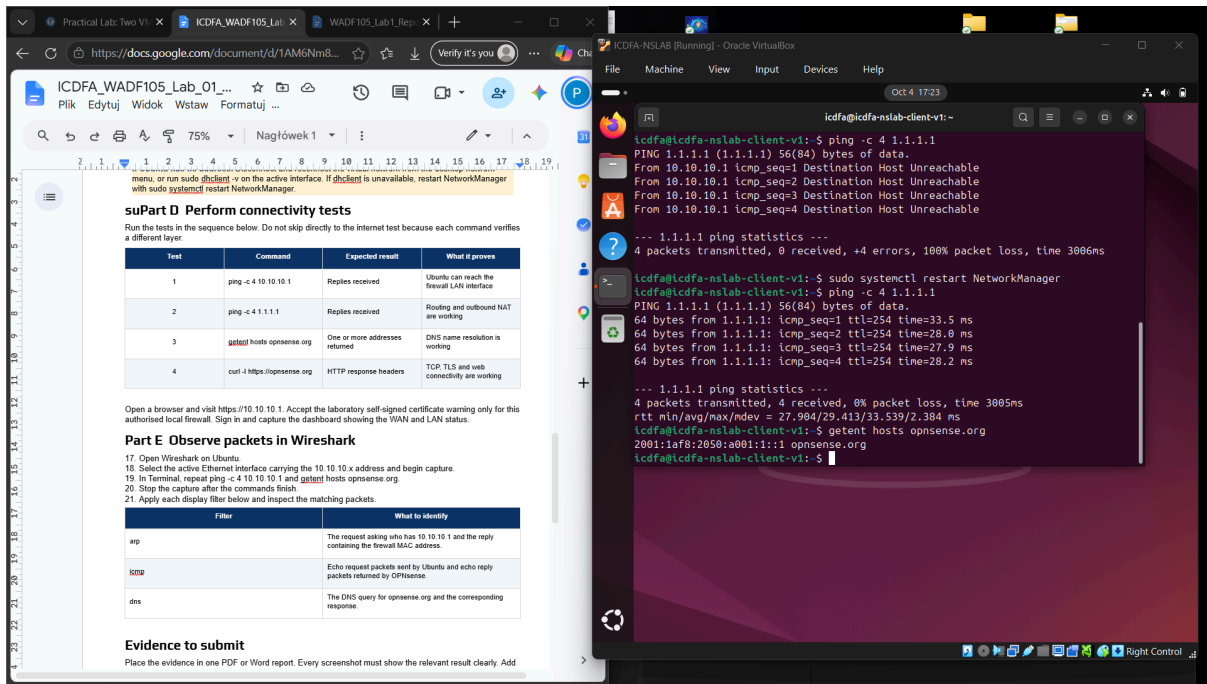




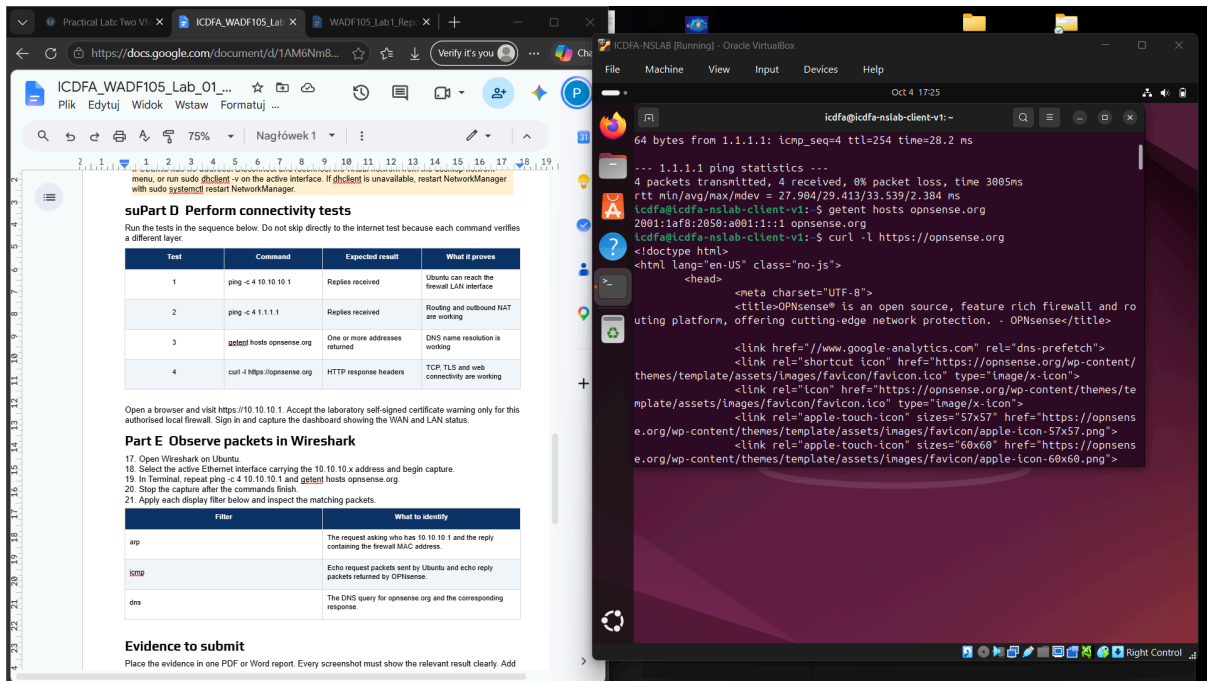
Success:

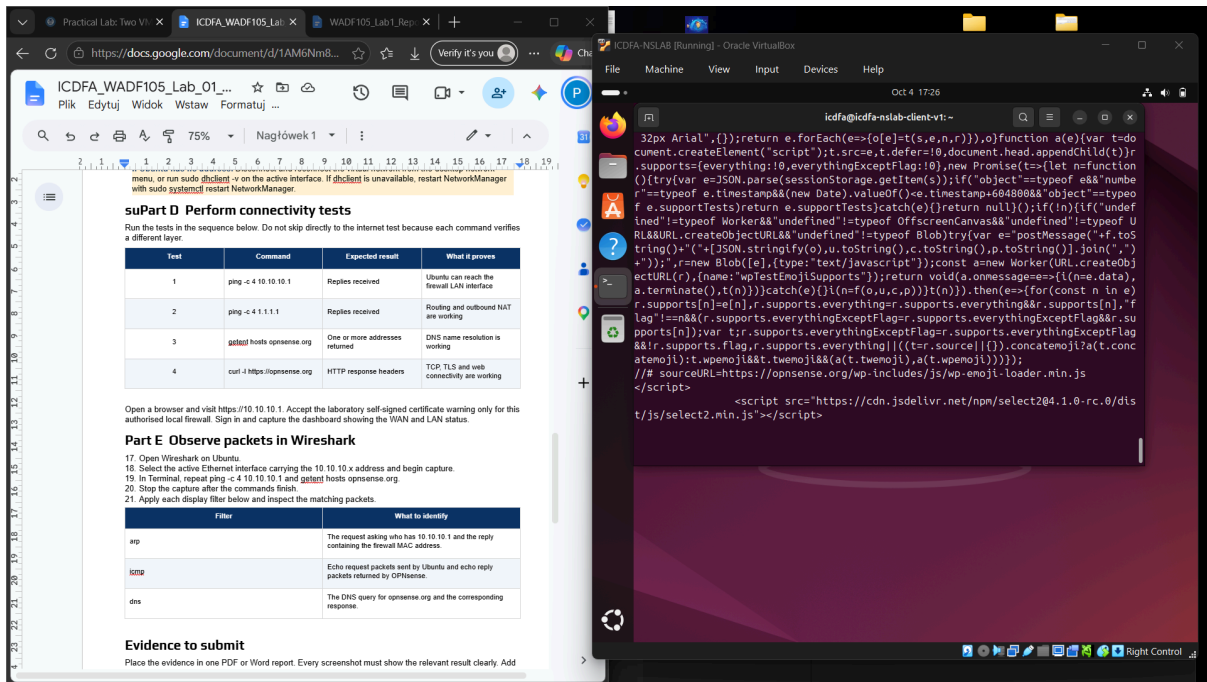


Command "getent hosts opsense.org" :

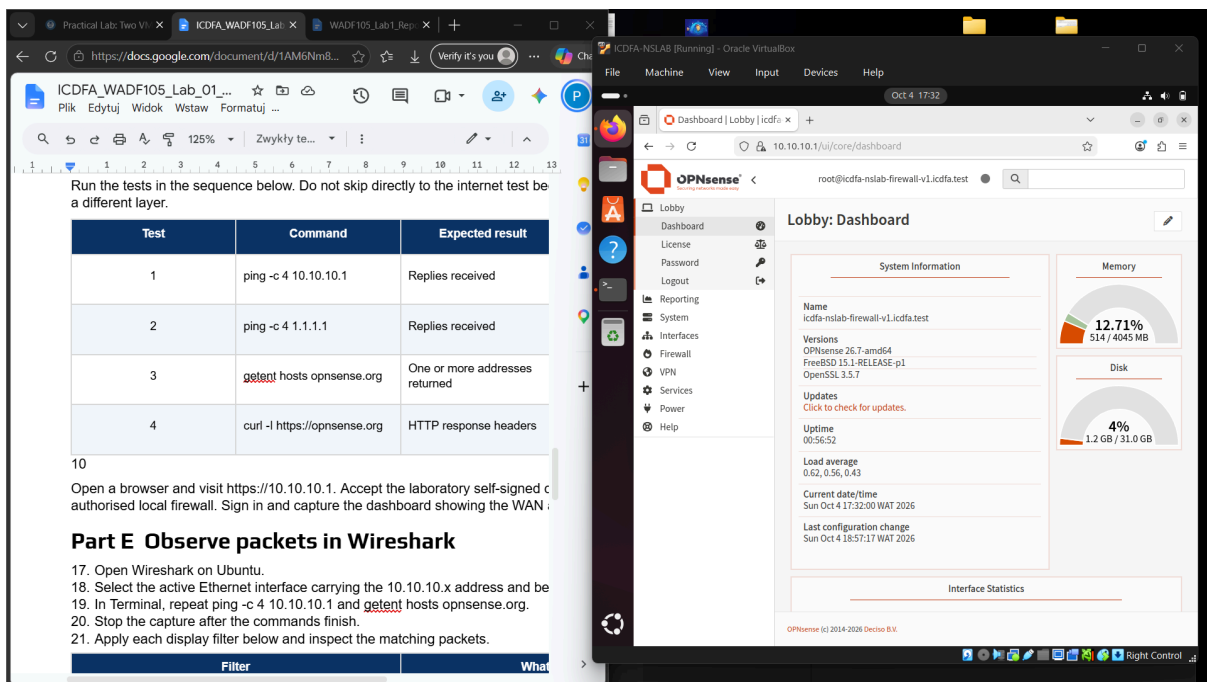


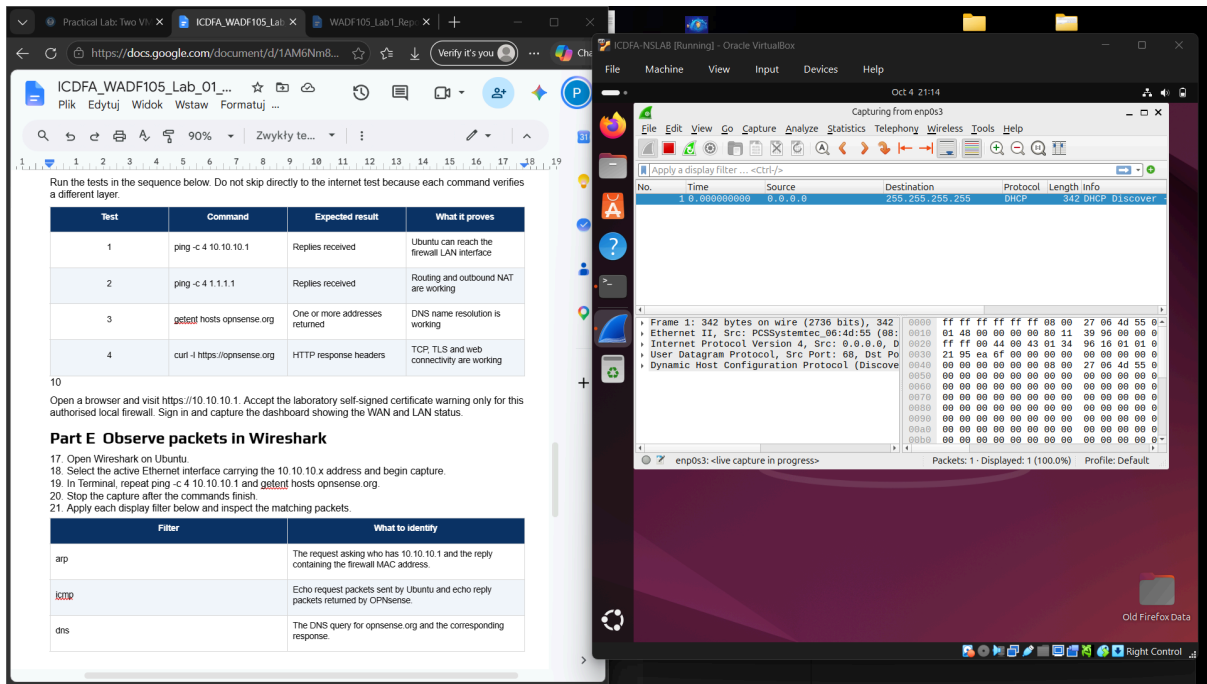
Command “curl -I https://opnsense.org”:



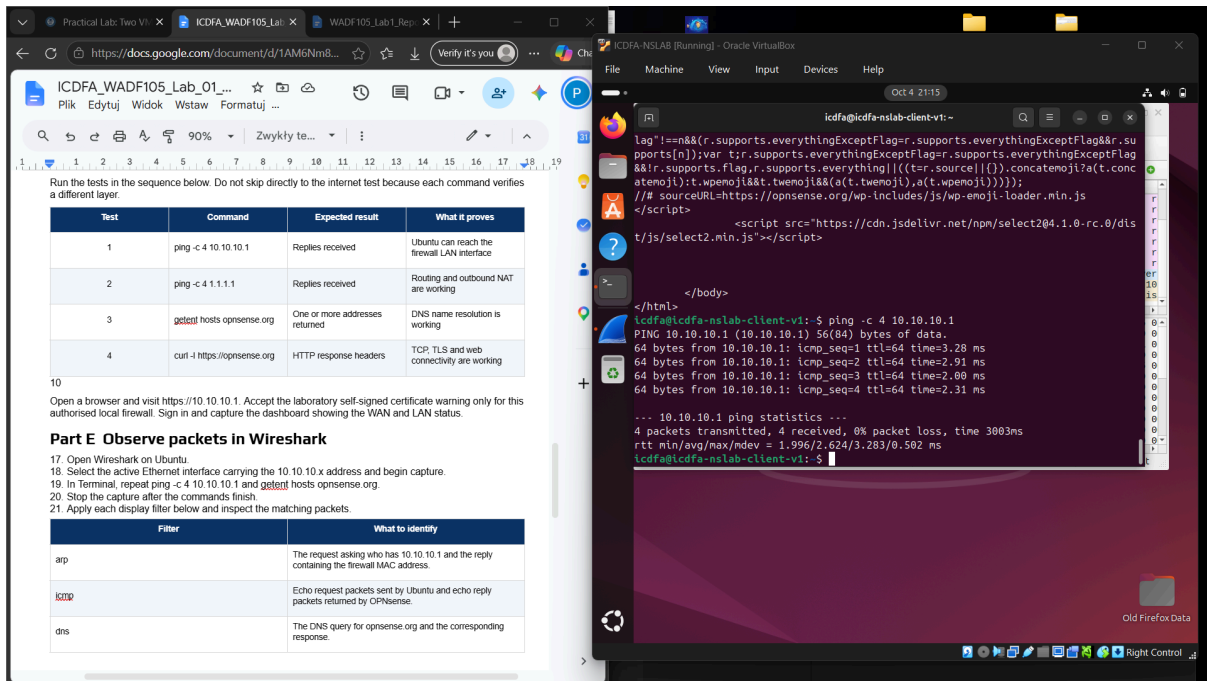


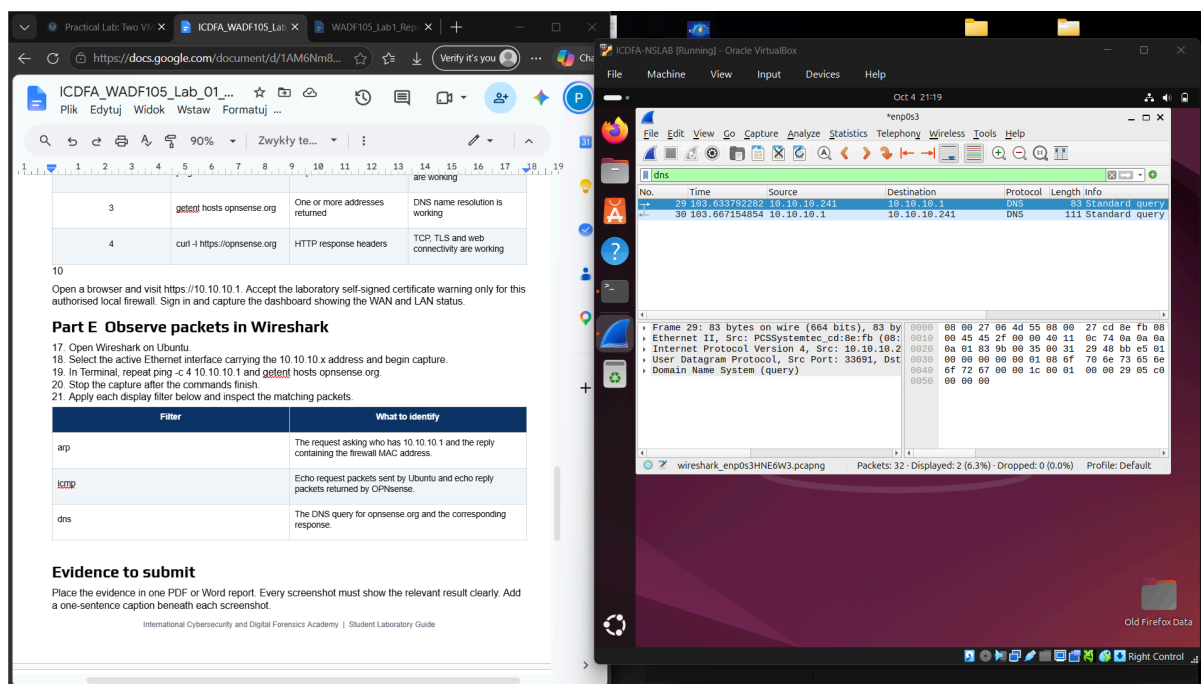
Opening OPNsense and recording Lan and Wan:





Testing command “ping -c 4 1.1.1.1”, “getent hosts opnsense.org” :





Questions answers:

1. What is the difference between the OPNsense WAN and LAN interfaces?

WAN is the interface connected to the outside world/internet through VirtualBox NAT, getting its IP automatically. LAN is the internal side connected to our private local network with a static IP (10.10.10.1), where OPNsense acts as the gateway and DHCP server for our client VM.

2. Why must both internal adapters use the same VirtualBox network name?

Because the internal network name acts like a virtual switch. For the firewall's LAN card and the Ubuntu client card to pass traffic back and forth at Layer 2, they both have to be plugged into the exact same named virtual network segment.

3. What information does the default route provide to Ubuntu?

It gives Ubuntu a default gateway (10.10.10.1). This tells Ubuntu where to send any traffic meant for IP addresses outside its own local subnet so the firewall can route it to the internet.

4. Which packet exchange allows Ubuntu to learn the firewall MAC address?

It's the ARP (Address Resolution Protocol) request and reply. Ubuntu asks "Who has 10.10.10.1?" and the OPNsense firewall replies with its actual MAC address so Ubuntu knows where to send frame headers.

5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

When you ping 1.1.1.1, you are using a raw IP address directly, which tests basic IP routing and NAT without doing a domain lookup. DNS is only tested when you try to resolve a domain

name (like opnsense.org) into an IP address. If DNS fails, 1.1.1.1 will still respond, but domain names won't load.